Nicholas Pianetto

[redacted]-[redacted]-[redacted] | [redacted]@gmail.com | pianetto.net

EDUCATION

Iowa State University, College of Engineering **Expected December 2026**

Bachelor Of Engineering, Computer Engineering

SKILLS

- **Programming Languages**: C, Java, Python, Verilog, VHDL
- **Hardware Tools**: Oscilloscope, Spectrum Analyzer, Soldering Iron (THT, SMT assembly and rework), Desoldering Gun, Multimeter, Variable Power Supply, High Voltage Discharge Probe, Logic Analyzer, Arduino, STM32, EPROM Programmer
- **Software**: IntelliJ, Visual Studio Code, QuestaSim, Quartus Prime, Android Studio, Git, Github, PADs Logic, PADs Layout, KiCad, STM32Cube IDE, Perforce, Lattice Diamond, FTDI Programmer (FTProg), Microsoft Suite, Linux

EXPERIENCE

Circuits II Teaching Assistant - ***Iowa State*** **January 2026 - Present**

- Provided constructive feedback for students in the lab on hands-on troubleshooting, equipment use, and circuit design using breadboards.

Research Assistant - ***Iowa State*** **May 2025 - Present**

- Collaborated with graduate students from the Animal Science department to design a bioimpedance analyzer for farm animals.
- Independently designed a schematic and board layout based around the ESP32, with working SPI SD data logging and LCD.

Embedded Design Engineer Intern - ***Ag Leader Technology*** **May 2024 - Dec 2024**

- Revised a serial to fiber optic PCB in PADs Layout and PADs Logic for use during ESD, EMC, and EMI testings.
- Developed a custom in-house inventory management application using Java (with MySQL) and Python (with SQLite), streamlining data handling.
- Collaborated with a team of embedded engineers to design the layout for a passive board in PADs while including proper EMI and ESD preventative design considerations.
- Performed board mods to prototype hardware by soldering experimental shielding and 0402 sized bypass capacitors in an effort to minimize capacitive coupling problems during ESD testing and found a solution to mitigate found issues.
- Independently developed the schematic and board layout for a passive USB-C PCB with impedance matching of all differential pairs to achieve support of DisplayPort functionality for product testing. Initiated and led design review of the schematic and layout.

Computer Museum Volunteer - ***Iowa State*** **March 2023 - Dec 2024**

- With support from the Director of Cyber Security at Iowa State, I helped with the curation of a computer museum using Iowa State's extensive collection dating back to the first digital computer, the Atanasoff-Berry Computer, from 1937.

PROJECTS

STM32 based ASCII to USB HID Controller

- Independently designed the schematic and layout, programmed, and assembled a custom PCB in KiCad based around the STM32F0 for use as a HID device that adapted a 47 year old 7-bit parallel ASCII keyboard to interface with modern devices.
- Successfully reverse engineered keyboard pinout without datasheets by using a logic analyzer and familiarity of 74LS logic.

PONG On Custom FPGA Development Board With Onboard VGA

- Independently designed (schematics and board layout), fabricated, and programmed a custom FPGA board based around Lattice Semiconductor's MachXO2 7000HC CPLD.
- Implemented support for 2 user-selectable voltage levels across different banks, pin headers for the MachXO2's IO, and 640x480 resolution VGA.
- Developed an FTDI FT2232H based JTAG programmer to interface with my board for easy plug and play development.